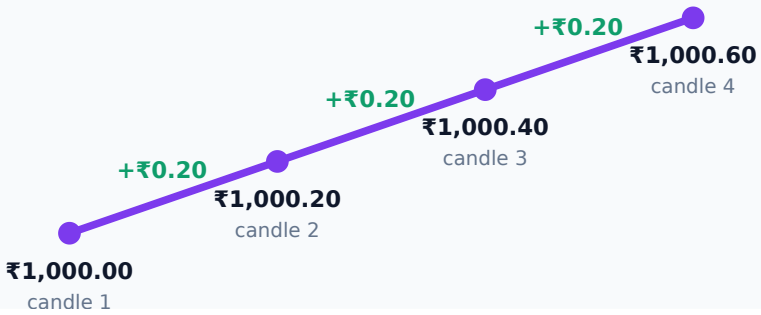
SMA 9 slope: Turn exit and V-turn

PalTra · Settings → 3. When to exit · simple guide with pictures. All examples: a ₹1,000 stock, your current numbers (steep run 0.0218, turn back 0.015, within 6 candles, turn exit 0.02, slope over 2 candles). The candles are made up but every SMA and slope number is worked out with the bot's own formula.

1. What "slope" means

Slope = how much the **SMA 9 line** moves per candle, as a % of the price. It is not an angle: the angle on a chart changes when you zoom, the % does not.



SMA 9 rises ₹0.20 each candle

₹0.20 ÷ ₹1,000 × 100

= slope +0.02% per candle

The bot compares the SMA 9 with 2 candles ago and divides by 2 ("Slope over" setting, default 2).

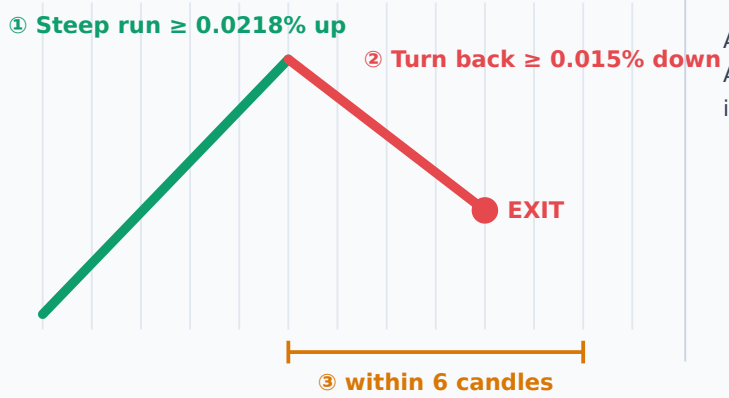
Same slope in rupees (how much the SMA 9 line moves per candle):

Slope	₹500 stock	₹1,000 stock	₹2,000 stock	₹5,000 stock
0.005% per candle	₹0.03	₹0.05	₹0.10	₹0.25
0.01% per candle	₹0.05	₹0.10	₹0.20	₹0.50
0.015% per candle	₹0.07	₹0.15	₹0.30	₹0.75
0.0218% per candle	₹0.11	₹0.22	₹0.44	₹1.09
0.02% per candle	₹0.10	₹0.20	₹0.40	₹1.00
0.03% per candle	₹0.15	₹0.30	₹0.60	₹1.50
0.05% per candle	₹0.25	₹0.50	₹1.00	₹2.50

Remember: the SMA 9 is an average of the last 9 closes, so it moves slower than the candles and a little late. One red candle hardly moves it; several candles in a row do.

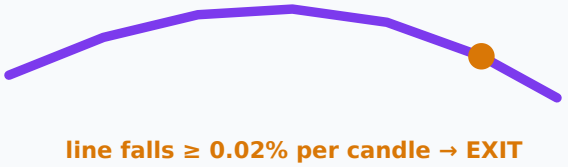
2. The three settings at a glance

V-turn (3 settings work together)



Turn exit (1 setting, alone)

Any shape. No steep run needed.
A slow rise and a slow drift down
is enough once the fall is steep enough.



Setting	Your value	What it means	On ₹1,000
① **Steep run** (with the trade)	0.0218% per candle	First the SMA 9 must have moved **this fast in your trade's direction** (up for a buy, down for a short).	line rising ≥ ₹0.22 per candle
② **Turn back** (against the trade)	0.015% per candle	Then the SMA 9 must slope **this much the other way**.	line falling ≥ ₹0.15 per candle
③ **Turn back within**	6 candles	A **count of candles**, not a slope: ② must happen within this many candles of ①. Later = not a V.	—
alone **Turn exit at**	0.02% per candle	A **separate rule**: close as soon as the SMA 9 slopes this much against the trade, whatever happened before. Works only with its **Turn exit** box ticked.	line falling ≥ ₹0.20 per candle

V-turn = ① + ② + ③ together. It catches a sharp V: fast up, then quickly down. A slow rounding top never counts.

Turn exit = one level only. It catches any fall that is steep enough, sharp or slow.

Turn entry (Settings → 1. When to enter) uses the **same** ① ② ③: a V down also opens a sell, a V up opens a buy. For a **short** everything is the mirror image: steep run = falling fast, turn back = rising.

Each picture on the next pages: candles with the SMA 9 line on top, the slope bars below with the three levels as dashed lines, and the yellow band = the 6-candle window after the last steep candle.

3. Example A: a sharp V, the V-turn fires

Bought at ₹1,000. Each candle then closes about **₹0.36 higher**, then about **₹0.96 lower**.



① On candle 6 the slope reaches **+0.0218%**: the steep run is there.

② On candle 10, only 4 candles later (inside the yellow 6-candle window), the slope is **−0.015% or more**: **V-turn → exit** (and a sell, with turn entry on). It fires on the 4th red candle: the SMA 9 kept rising for 2 red candles because it lags.

③ The turn exit (−0.02%) would fire one candle later, on candle 11, only if its box is ticked.

Candle	Close	Move	SMA 9	Slope %/candle	What happens
1	₹1,000.36	+0.36	₹1,000.04	+0.0020	buy
2	₹1,000.72	+0.36	₹1,000.12	+0.0060	
3	₹1,001.08	+0.36	₹1,000.24	+0.0100	
4	₹1,001.44	+0.36	₹1,000.40	+0.0140	
5	₹1,001.80	+0.36	₹1,000.60	+0.0180	
6	₹1,002.16	+0.36	₹1,000.84	+0.0220	**steep run reached**
7	₹1,001.20	−0.96	₹1,000.97	+0.0187	
8	₹1,000.24	−0.96	₹1,001.00	+0.0080	
9	₹999.28	−0.96	₹1,000.92	-0.0027	
10	₹998.32	−0.96	₹1,000.69	-0.0153	**V-turn → exit**
11	₹997.36	−0.96	₹1,000.32	-0.0300	**turn exit**
12	₹996.40	−0.96	₹999.80	-0.0446	

4. Example B: a slow rounding top, only the turn exit fires

Bought at ₹1,000. A slow climb of about **₹0.15 per candle**, then a slow drift down of about **₹0.42 per candle**.



① The slope never reaches **+0.0218%** (best about +0.0125%): **no steep run, so no V**. The V-turn never fires, however far it falls.

③ The **turn exit** does not care about the past: on candle 15 the slope is **−0.02% or more**, so it closes the trade (only with the Turn exit box ticked). Without it, this trade waits for the stop or another exit.

Table: from candle 6 on.

Candle	Close	Move	SMA 9	Slope %/candle	What happens
6	₹1,000.90	+0.15	₹1,000.35	+0.0092	
7	₹1,001.05	+0.15	₹1,000.47	+0.0108	
8	₹1,001.20	+0.15	₹1,000.60	+0.0125	
9	₹1,000.78	−0.42	₹1,000.69	+0.0110	
10	₹1,000.36	−0.42	₹1,000.71	+0.0055	
11	₹999.94	−0.42	₹1,000.67	-0.0008	
12	₹999.52	−0.42	₹1,000.57	-0.0072	
13	₹999.10	−0.42	₹1,000.40	-0.0135	
14	₹998.68	−0.42	₹1,000.17	-0.0198	
15	₹998.26	−0.42	₹999.88	-0.0262	**turn exit**
16	₹997.84	−0.42	₹999.52	-0.0325	
17	₹997.42	−0.42	₹999.10	-0.0388	
18	₹997.00	−0.42	₹998.68	-0.0420	

5. Example C: the turn back comes too late, no V

Bought at ₹1,000. A steep climb (₹0.36 per candle), then **14 flat candles**, then a fall of about ₹0.95 per candle.



① The steep run is there on candle 6.

② The fall comes long after the yellow 6-candle window has ended, so it is **not a V**: the V-turn does not fire. (Raise "Turn back within" to catch slower V's.)

③ The turn exit still fires on candle 23, when the slope is −0.02% or more (if ticked).

Table: candles after the flat stretch only.

Candle	Close	Move	SMA 9	Slope %/candle	What happens
19	₹1,002.16	±0.00	₹1,002.16	+0.0000	
20	₹1,002.16	±0.00	₹1,002.16	+0.0000	
21	₹1,001.21	−0.95	₹1,002.05	-0.0053	
22	₹1,000.26	−0.95	₹1,001.84	-0.0158	
23	₹999.31	−0.95	₹1,001.53	-0.0263	**turn exit**
24	₹998.36	−0.95	₹1,001.10	-0.0369	
25	₹997.41	−0.95	₹1,000.58	-0.0474	
26	₹996.46	−0.95	₹999.94	-0.0580	

6. Your settings and how to change them

Right now: Turn entry is **ON** (uses ① ② ③), Turn exit is **off** (its number does nothing), V-turn exit is **off**. So a V against an open trade closes it and opens the other way through the turn entry. Your trade R814-24 (02 Nov, bought 1,837.80 at 12:45 on an SMA 9 turn, closed 1,824.00 at 13:15 "SMA 9 turned against") was this: small V's in a flat, tangled stretch where the SMAs had no direction.

If you want…	Change
Fewer, cleaner V's (fewer whipsaws)	Raise ① **Steep run** (e.g. 0.03) and ② **Turn back** (e.g. 0.025).
Only quick, sharp V's	Lower ③ **Turn back within** (e.g. 4).
Also catch slower V's	Raise ③ **Turn back within** (up to 10).
A smoother slope, less noise	**Slope over** 3 candles instead of 2, or read the slope on a bigger **slope candle**.
Close on any steep enough fall, V or not	Tick **Turn exit** and set **Turn exit at** (e.g. 0.02).
No trade opened on a V that points away from the 9/21 cross	Tick **Only toward the cross** (1. When to enter).
Hold until a candle of the other colour confirms	Tick **Close on a candle of the other colour**.

Before LIVE: change one number at a time and replay a few past days on the backtest server, then compare the runs in the Backtests tab. The Settings page's slope box also shows these example candles live for any price you type.

Formula the bot uses: slope = (SMA 9 now − SMA 9 two candles ago) ÷ SMA 9 then × 100 ÷ 2, in % per candle, read once each candle closes.